

Prologue

Programming and Understanding

One way to become aware of the precision required to unambiguously communicate a mathematical idea is to program it for a computer. Rather than using canned programs purely as an aid to visualization or numerical computation, we use computer programming in a functional style to encourage clear thinking. Programming forces us to be precise and unambiguous, without forcing us to be excessively rigorous. The computer does not tolerate vague descriptions or incomplete constructions. Thus the act of programming makes us keenly aware of our errors of reasoning or unsupported conclusions.¹

Although this book is about differential geometry, we can show how thinking about programming can help in understanding in a more elementary context. The traditional use of Leibniz's notation and Newton's notation is convenient in simple situations, but in more complicated situations it can be a serious handicap to clear reasoning.

A mechanical system is described by a Lagrangian function of the system state (time, coordinates, and velocities). A motion of the system is described by a path that gives the coordinates for each moment of time. A path is allowed if and only if it satisfies the Lagrange equations. Traditionally, the Lagrange equations are written

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0.$$

What could this expression possibly mean?

Let's try to write a program that implements Lagrange equations. What are Lagrange equations for? Our program must take a proposed path and give a result that allows us to decide if the path is allowed. This is already a problem; the equation shown above does not have a slot for a path to be tested.

¹The idea of using computer programming to develop skills of clear thinking was originally advocated by Seymour Papert. An extensive discussion of this idea, applied to the education of young children, can be found in Papert [13].